“NVIDIA Builds Out Its Omniverse Ecosystem to Support the Automotive Metaverse”

— SiliconANGLE

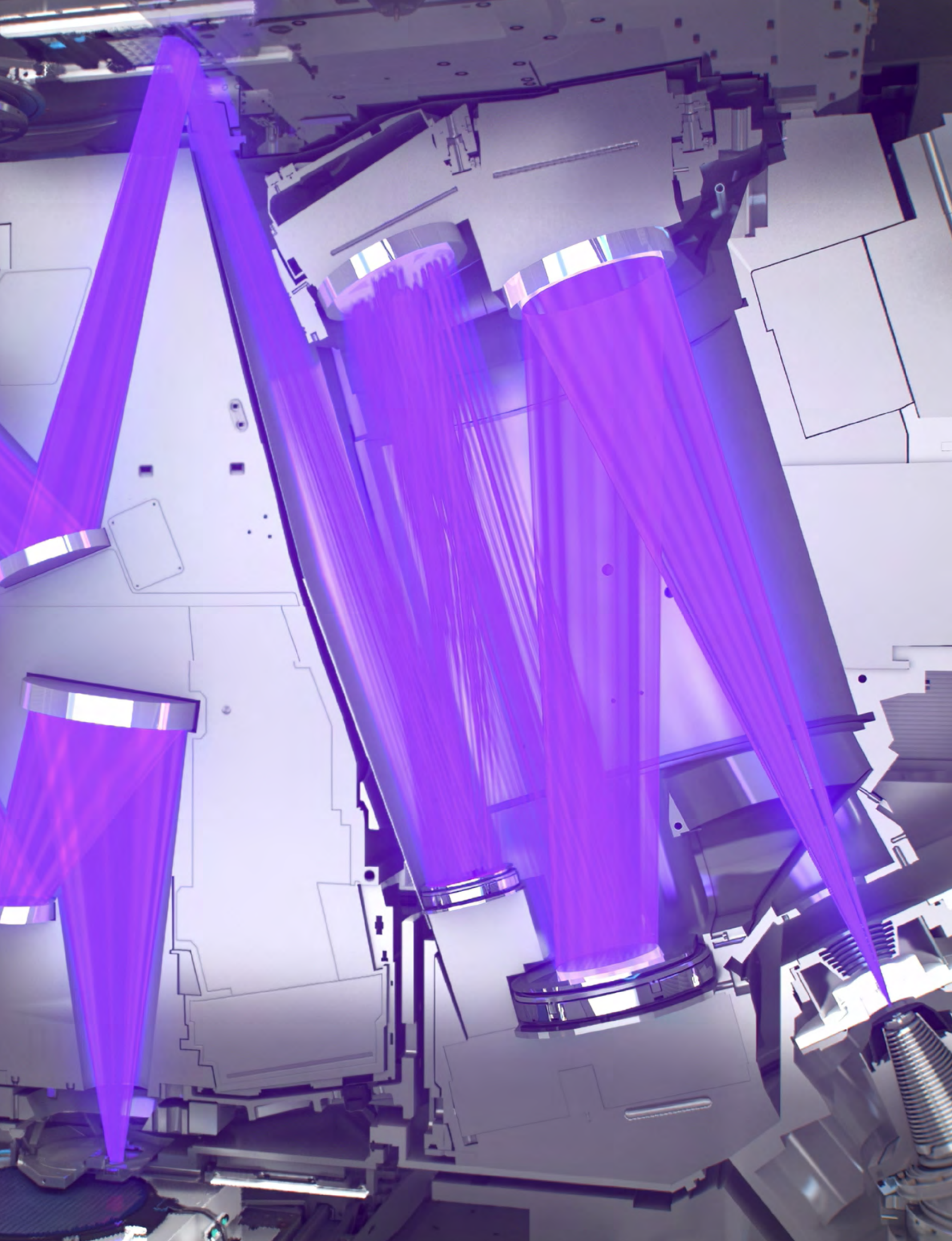
NVIDIA Omniverse is an ideal tool for an industrial world seeking to digitalize. Omniverse can simulate the best possible layouts before the first brick is placed. The $3 trillion automotive industry is modernizing all its processes to take advantage of computing and AI. BMW Group uses Omniverse to build a whole factory digital twin before constructing it physically. Mercedes-Benz is using NVIDIA DRIVE IX on Omniverse to design and simulate its integrated cabins and electronics.

“NVIDIA cuLitho Computational Lithography Massively Accelerates Chip Design Using GPUs”

— Forbes

NVIDIA’s acceleration libraries solve new challenges and open new markets. They connect to applications that connect to the world’s industries, forming a network of networks.

NVIDIA cuLitho is a new library that supercharges computational lithography, an immense computational workload in chip design and manufacturing. The result of over four years of partnership and collaboration with TSMC, ASML, and Synopsys, cuLitho accelerates computational lithography by over 40X and paves the way for the industry to go to 2nm and beyond.



Dear NVIDIANs and Stakeholders,

We had a tough 2022. Our business was affected by economic headwinds, geopolitical tension, and a product supply chain that swung from severe shortage to excess. NVIDIANs rose to tackle each challenge while inventing new technologies and capabilities that position us at the center of the most exciting opportunities in the history of computing.

Data center AI and accelerated workloads are continuing to skyrocket. Developers are shifting to NVIDIA accelerated computing as the four-decade-long exponential scaling of CPU-based general-purpose computing ends.

ChatGPT, the AI application heard around the world, showcased the abilities of generative AI and its potential to drive industrial productivity and advance the world's most significant scientific challenges. Generative AI has created a sense of urgency in companies everywhere to reimagine their products and business models.

Generative AI is also accelerating industrial digitalization. The largest industries, from auto manufacturing to pharmaceutical, will be reinvented with generative AI and become some of the most advanced technology industries.

NVIDIA's body of work is reshaping the future of computing.

Accelerated Computing …

Accelerated computing is not easy—it requires full-stack optimization from chip architecture, systems, and acceleration libraries, to refactoring the applications. Each application domain requires optimized stacks—from graphics, imaging, and particle or fluid dynamics to data processing and machine learning. NVIDIA pioneered accelerated computing by extending the GPU, a 3D graphics accelerator, into a parallel computing accelerator.

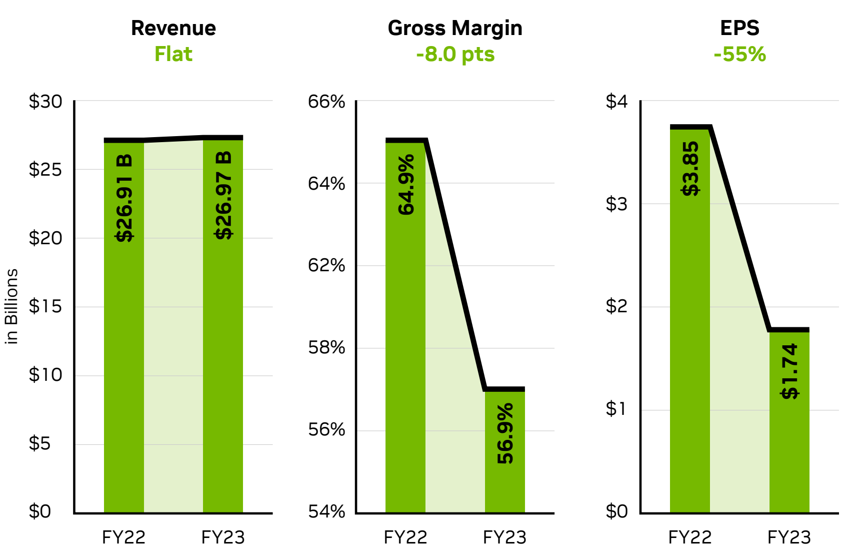
NVIDIA acceleration libraries build on CUDA, and all NVIDIA GPUs are CUDA-compatible. Four million developers are working with CUDA, and that number is expanding. It took 12 years to reach 2 million developers, but we've doubled that number in the last two and a half years. CUDA has been downloaded more than 40 million times.

A wealth of accelerated applications attracts end users, which creates a large market for cloud service providers and computer makers to serve. This in turn affords billions in R&D to fuel its growth. NVIDIA has established the accelerated computing virtuous cycle.

We help developers achieve incredible speedups through full-stack invention, from the chips and systems to the algorithms and applications they run. These algorithms are optimized and packaged into acceleration libraries, where they help millions of developers across industries solve complex problems. NVIDIA has hundreds of acceleration libraries that form our core platforms: NVIDIA RTX for graphics, NVIDIA HPC for scientific computing, NVIDIA AI for data science and artificial intelligence, NVIDIA DRIVE for autonomous vehicles, and NVIDIA Omniverse for industrial digitalization applications.

We now offer 300 acceleration libraries and 400 AI models, with 100 added or updated in the past year alone, including cuQuantum for quantum computing, cuOpt for combinatorial optimization,

GAAP Results



and the new cuLitho, which addresses the single largest computation workload in chip design and manufacturing: computational lithography. The product of close collaborations for over four years with TSMC, ASML, and Synopsys, cuLitho can reduce mask creation time from two weeks to overnight and enable 500 NVIDIA DGX H100 systems to achieve the same work as 40,000 CPU systems. That's 1/8 of the space needed and 1/9 of the power to achieve current results with NVIDIA systems. Most importantly, it provides the chip industry a platform for the next leap into the next chip design and manufacturing miracle. It operates near the limits of physics: high-numerical aperture extreme ultraviolet lithography. This process will require radical algorithms such as inverse lithography and AI approaches for continued scaling to 2nm and beyond.

… Is Sustainable Computing

The exponential growth trend in CPU computing performance coupled with only moderate power and cost increases is no more. Data centers are already about 1-2% of global electricity consumption and that consumption is expected to continue to grow. This continued growth is not sustainable for operating budgets and our planet.

For computing to be sustainable, data centers must accelerate every workload possible. NVIDIA pioneered accelerated computing and has built a large installed base and a rich ecosystem of developers and applications available everywhere. Accelerated computing is sustainable computing, bending the energy consumption trendline away from runaway growth and toward a sustainable future.

The iPhone Moment For AI

ChatGPT has taken the world by storm. Hundreds of millions of people worldwide have been captivated by its abilities. Companies in new and established industries are racing to explore its potential.

The foundation of ChatGPT is a large language model (LLM). According to ChatGPT itself, an LLM is:

"A large language model refers to a type of artificial intelligence (AI) model trained on a vast amount of text data to generate human-like text responses. Language models understand and generate human language, and they can perform a wide range of tasks, such as answering questions, summarizing text, translating languages, and generating text content.

"LLMs, such as GPT-3 (Generative Pre-trained Transformer 3), are known for their high capacity to learn from large datasets and generate coherent and contextually relevant text. These models use deep neural networks with multiple layers to capture complex patterns and representations in language data. LLMs can understand the nuances of language, including grammar, syntax, context, and meaning, and can generate text that is often difficult to distinguish from text written by humans.

"LLMs have many potential applications in various domains, including natural language processing (NLP), conversational AI, content generation, language translation, and more."

LLMs can learn the representation of information that has structure from human language, music, images, and even proteins and chemicals. The potential of generative AI models to amplify human productivity is incredible. Impacts will be felt in industries ranging from healthcare and financial services, to design, art, and entertainment. LLMs are trained with NVIDIA DGX AI supercomputers. An LLM neural network model with tens of billions of artificial neurons learns by processing trillions of bytes of data. This requires